

Speed-crazy USB 3.0 promises backward compatibility, too

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■ ABHIJIT PAUL CHOUDHURY

It's a dream come true for the speed-crazy people. With a superior capability of transferring data, enters the SuperSpeed, also known as USB 3.0—a major advancement in ubiquitous technology. After almost a decade since USB 2.0's inception in 2000, USB is poised for its first major upgrade in years.

With its ten times faster speed and other new features, SuperSpeed would leave the existing USB 2.0 far behind, when it is launched this year-end. While USB 2.0 offers speed of 480 megabits per second, USB 3.0 jumps to 5 gigabits per second. A revolution indeed!

For today's time-crunched consumers, who need to transfer large digital files fast from PCs to devices and vice-versa, USB 3.0 not only promises faster speed but also backward compatibility. More than one billion products in the market conform to USB 2.0 specifications. Keeping this in mind, the new USB 3.0 specifications have been made backward compatible so that these products don't get defunct. The Super-

Will Windows & Linux support it?

Windows or Linux operating systems available till date don't support USB 3.0 specifications. But once the host controllers and the devices are available in the market, the operating system vendors will provide patches for their respective operating systems. By updating the operating system with these patches provided by Windows or the packages provided by Linux, the older operating systems will be able to support the USB 3.0 specification.

USB Speeds

Version	Speed
USB 1.0	1.5 Mbps
USB 1.1	14 Mbps
USB 2.0	480 Mbps
USB 3.0	5 Gbps

Speed is also considered to be power-economical. Hence, it meets the needs of everyone—from the tech-savvy executive to the average home user.

Due to its super-fast speed, USB 3.0 requires fibre-optic cabling and hence adds a length of optical data cable to the mix, retaining its full compatibility with both USB 2.0 and USB 1.0. The cable, the host device (for example, the PC) and the peripheral device all must be USB 3.0 capable to achieve the 3.0 speeds, otherwise, the connector reduces to the lowest common denominator—the USB 2.0 speed.

Recently, at a consumers electronics show, Symwave announced its collaboration with Seagate to demonstrate Symwave USB 3.0 storage controller

devices designed to comply with SuperSpeed specifications. Universal Serial Bus Implementation Forum (USB-IF), a non-profit organisation, supports and promotes USB, wireless USB and USB On-the-Go and takes care of the specifications' maintenance.

According to Jeff Ravencraft, president and chairman of USB-IF, USB discrete controllers will be available in the second half of this year and consumer products are estimated to appear in the first half of 2010 with broad development throughout the year.

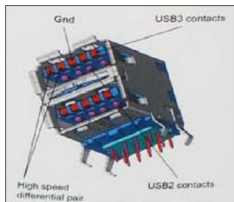
"The devices that will take advantage of SuperSpeed USB are likely to include data-storage devices like flash drives, hard drives, solid-state drives, digital cameras and music and video players. As adoption becomes more widespread, more SuperSpeed USB products will appear in the market," adds Ravencraft.

Jan Axelsson, author of USB Complete, says, "The first device to take advantage of USB 3.0 will be a mass storage device, and videos are likely to be another application for USB 3.0."

USB 3.0 is capable of efficiently handling power. It does no device polling, so less power is drained, which was not the case in USB 2.0. USB 3.0 also provides more power to its devices, hence devices connected to it will get charged faster. Configured devices will receive 900 milliamperes and suspended devices will receive 150 milliamperes in USB 3.0 as compared to 500 milliamperes and 100 milliamperes, respectively, in USB 2.0.

The unique features put USB 3.0 ahead of eSATA (3Gbps)—the only other device which can give it a fitting challenge. It remains to be seen whether the combination of USB 3.0 and eSATA will push FireWire out of the market completely. ■

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Arrangement of USB 3.0 and USB 2.0 contacts